

though. As more people store their photos and documents online, it is only a matter of time when editing them online will feel as natural.

Gaining parity

Imagine an online image editor created a decade ago. Such an “application” would not have been able to read a local image, or manipulate it in the local browser. At most the user could select an image, upload it to the server, specify some operations to perform, and get back an edited image to download. Also such an “application” would not work offline, and would be terribly slow if your images were even a few MB. It is obvious why no one would use such an app.

Now look at how things stand today. The new File APIs will allow a web app to open and edit local files, and the Canvas API allows one to manipulate graphics making it possible to edit images in the browser itself. This is also helped by the fact that JavaScript is fast enough to do this in the browser now. Using new specifications for offline applications, a web app can be designed to work even when there is no connectivity. Other than the fact that it is running in the browser, this is quite similar to any native application.

It is unlikely that web based applications will ever reach the same parity in performance or features as native applications, however they can offer a different set of advantages that can weigh the scales in their favour. Web applications are easier to develop and update, and all your users run the latest version. There is no need to install the software, or uninstall it, just open the site and it works.

The web as a platform is becoming increasingly powerful as new standards continue to be proposed based on actual needs of developers. You can see a list of some of the things that you can achieve using new web standards in the sidebar.

Native advantage?

Is there really an advantage to go native now? Sure performance is a concern if your application requires heavy processing. Or is it? Consider YouTube, each uploaded video is converted to a plethora of formats that are to run on all manner of devices. Many YouTube users are uploading from devices that aren't even powerful enough to encode video, so YouTube does it for them. Another example is Autodesk Homestyler that lets you model houses in 3D in your browser; these can then be rendered in high quality by Autodesk servers so they can

“A number of platforms are now embracing web standards as a means to create applications”

*pointers

>>Nuggets of cool code at work.



*Windows 8 UI for tablets

>>David Washington, Sr Program Manager, Windows User Experience Team, Microsoft demos the Windows 8 UI for tablets
Duration > 01:21

<http://dvw.in/dec-11-p1>



*Improved search in Windows 8

>>Brian Uphoff, Program Manager, Search, View, and Command Team shares new search functions > 02:45

<http://dvw.in/dec-11-p2>



*jQuery Mobile

>>The first update for jQuery Mobile 1.0 has been released. Here, Raymond Camden gives an introduction to jQuery Mobile.
Duration > 46:22

<http://dvw.in/dec-11-p3>



*JDK7 in a nutshell

>>At O'Reilly OSCON Java 2011, Joe Darcy simplifies Java 7. Here's the larger picture of how Java has changed from JDK 1.2 through the years, to where it is today.

Duration > 09:34

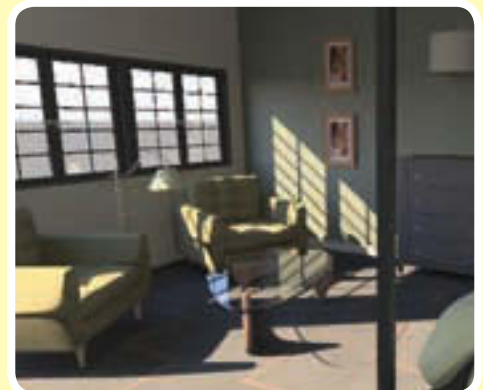
<http://dvw.in/dec-11-p4>

be viewed even by users having systems too slow to do the rendering. As you can see, in these cases, the performance of a web application can be significantly more than that of a local app. You will find that most disadvantages have a counter advantage.

We have now grown used to the concept of offline apps, or apps that don't need an internet connection to work. However even native applications are increasingly becoming internet aware, and thus internet dependent. Whether it is for activation or updates, or even for downloading the application in the first place! Internet is becoming a requirement.

Today an average user will browse the internet for an application that fits their need, download it, install it, and finally run it. If it costs, then you have to pay for it and probably activate it as well. This process may need to be repeated for each update or upgrade.

Now compare this to a web application. You browse the web for an app that matches your needs, you open the app, and it just works! If you need to create an account, that is an added one time process, and payments are usually recurrent with no need to



>>A 3D scene rendered by Autodesk Homestyler. All possible on a web browser with realistic textures and effects

pay for upgrades. It is actually more convenient to open the website since downloading and “installing” the software isn't required. Add to this the fact that you can do this whether you are running Windows, Linux, Mac OSX, iOS, Android, BlackBerry or any esoteric OS running on any esoteric device as long as it supports a web standard capable browser, you don't need to pay again for a license. In the end, the process of using a web application is also one of downloading (your browser does this, and if supported it caches it), “installing” (or creating a bookmark), and then launching (via the bookmark or URL).

Today if you are looking to create an application that will reach mobile, tablets, desktops, and perhaps even TVs, there is one “runtime” / platform you could consider developing for; and that is your browser. >

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